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Gunshot Residues and Metals
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INTRODUCTION

The development of a method for the identification of gunshot residue (GSR) on the hands of a suspected shooter has been a top priority of forensic chemists for many years. Law enforcement officers and prosecutors have good reason to believe that placing the "smoking gun" in the hands of a suspected shooter by means of an incontrovertible scientific test would provide the trier of fact with a significant element in the proof required for a conviction. From a cursory study of the history of GSR tests, it is clear that such a scientific test does not exist (FBI Law Enforcement Bulletin 1935, 1940; Harrison and Gilroy 1959; Ruch *et al.* 1964; Cowan *et al.* 1973).

The tests that are in use in forensic laboratories today are either quantitative tests for chemical elements in ammunition primer mixtures or a test which specifically identifies primer residue particles by scanning electron microscopy and X-ray analysis (Wolten *et al.* 1977). None of these tests addresses the issue of how the primer residue is deposited on the hands. Nevertheless, the GSR tests are still valuable provided their limitations are understood.

Specimens taken from the hands of suspected shooters and submitted to the FBI Laboratory are analyzed by neutron activation analysis (NAA) for the elements antimony and barium. These elements are components of most primer mixtures and can be deposited on the hands when a firearm is discharged. In the early 1960s, the group at Gulf General Atomic (San Diego, CA) demonstrated that NAA was an excellent method for measuring these elements accurately over the wide concentration range found on the hands of nonshooters and shooters. These amounts typically vary from a few nanograms

to several micrograms (Ruch *et al.* 1964). Their work showed that in many situations there was an increase in the amounts of antimony and barium on the back of the hand when a firearm was discharged compared to the amounts of these elements normally on the hands (Schlesinger *et al.* 1970). Based on this introductory work, the availability of a research nuclear reactor in the Washington, D. C. area and our own considerable experience, NAA has been the analytical method of choice for the examination of GSR samples for antimony and barium since the inception of the FBI Laboratory's GSR program in the late 1960s. This method is used despite the availability of the ostensibly less costly flameless atomic absorption spectrophotometry (FAAS) method used by many crime laboratories for GSR analysis. In our view, the analytical reliability of the NAA method for quantifying antimony and barium over a wide range of concentrations, coupled with the ability to process a high volume of samples through the use of a computer-controlled sample changer and automatic data reduction, make the FBI Laboratory's GSR program a cost-effective one.

SCIENTIFIC PRINCIPLE

The basis for the NAA method is to produce radioisotopes in a material by bombarding the material with neutrons. The stable isotopes of some of the elements in the material are transformed into radioisotopes. Many of these radioisotopes decay with the emission of gamma rays. The energy of the gamma rays and the rate of radioactive decay (half-life) are used to identify the radioisotopes of interest produced in the material. The intensity of the gamma rays of a specific energy is a direct measure of the concentration of an element in a material. In the case of the analysis of GSR samples for antimony and barium, the radioisotopes measured are 83 minute ¹³⁹Ba (166 keV photopeak) and 67.2 hour ¹²²Sb (564 keV photopeak).

ANALYTICAL PROCEDURE

The FBI Laboratory examines cotton swabs submitted by Federal, state and local law enforcement agencies in criminal matters as well as swabs from test firings and handblank projects. Generally, these swabs are of the cotton applicator type which have been moistened with a few drops of 5 percent nitric acid and applied to the hand. Each hand must be swabbed, and moistened swabs which have not been applied to any surface must also be provided as a materials blank. After a microscopic examination to determine the condition and degree of utilization of the swabs, they are removed from their sticks and packaged in labeled polyethylene bags. These bags are loaded into an irradiation container called a rabbit. The rabbit holds swabs from approximately ten persons (right hand, left hand, blank) and three vials of liquid standards, each vial containing enough solution to provide 1 μg barium and 0.1 μg antimony in 100 μl . The swabs and standards are irradiated for 1.5 minutes at a thermal neutron flux of approximately 10^{14} n/cm²/sec in the National Bureau of Standards Reactor located in Gaithersburg, Md.

A post irradiation chemical separation is performed for two reasons. One is to remove some of the nonpertinent radioisotopes formed during the irradiation thereby reducing the total radioactivity of each sample. The second is to prepare a reproducible sample configuration to present to the radiation detector. (Even though antimony and barium can often be measured without a chemical separation, the condition and shape of the swabs make quantitation unreliable.)

CHEMICAL SEPARATION

The irradiated swabs from each sample are placed in a 100 ml disposable polyethylene beaker. A carrier solution (80°C, 25 ml) is added to each beaker. This solution contains 20 mg of barium as barium nitrate and 20 mg of antimony as antimony

potassium tartrate in 3N hydrochloric acid. The cotton is mechanically dispersed, allowed to leach for at least 5 minutes to permit isotopic exchange, and then filtered on Whatman #4 paper. The filtrate is collected in a second disposable beaker. After washing, the final volume is approximately 70 ml.

Sodium sulfate (10 percent, 5 ml) is added to the filtrate to precipitate barium as barium sulfate. At least 5 minutes are required for nearly complete precipitation of the barium sulfate. Sodium sulfide (10 percent, 5 ml) is then added to precipitate antimony as antimony sulfide. Nearly complete formation of this precipitate also requires about 5 minutes. The mixture of precipitates is filtered on 47 mm 0.22 μm porosity Millipore filter paper (Figure 1). The filter paper is placed in a labeled 47 mm disposable petri dish for automatic gamma counting.

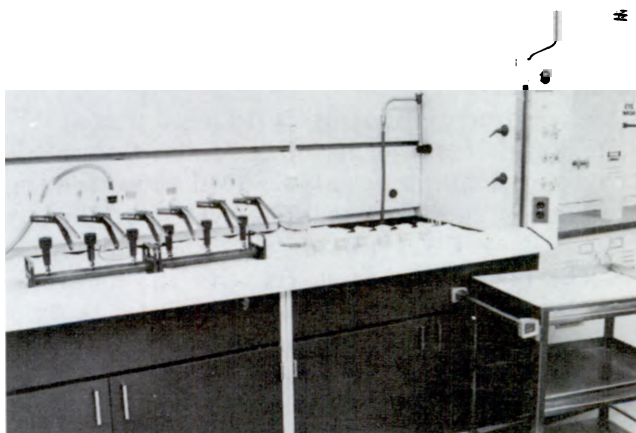


FIGURE 1. Manifold setup for filtration of antimony and barium precipitates.

Chemical yields for the antimony and barium in solution as determined using ¹²⁵Sb and ¹³³Ba radioactive tracers approach 100 percent. It is pointed out that antimony in some swabs does not completely exchange with the carrier. This has been determined by performing gamma counting on the swab materials after the separation procedure. (Generally, the antimony retained in the swabs does not exceed 10 percent of the total.)

The three standards are processed the same way except that the first filtration is eliminated.

GAMMA COUNTING AND DETERMINATION OF ANTIMONY AND BARIUM CONCENTRATIONS

When analyzing numerous samples, the time spent acquiring a spectrum for each sample is limited by the half-life of the radioisotope being measured. Acquiring a spectrum, a process called "counting," for too brief a period will yield insufficient data for a meaningful analysis. Allowing too much time for each sample will result in the loss of data from the short-lived radioisotopes in the last samples to be counted. Since barium decays much more rapidly than antimony, the samples are counted twice. In the first count, performed to determine the mass of barium in the samples, the counting time for each is typically less than 2 minutes. After the barium results are printed, the samples are counted again for a much longer period of time, in order to measure the longer-lived antimony component.

The gamma counting is done on a coaxial lithium-drifted germanium detector coupled to a multichannel analyzer (MCA) with data reduction capabilities. The system resolution is 1.95 keV FWHM using 1.33 MeV photons.

The barium count starts with the loading of the petri dishes onto an automatic sample changer (Figure 2). From this point on, the manipulation of each sample and the acquisition, storage and analysis of each spectrum are controlled by the MCA. The MCA begins the count by causing the sample changer to move the first sample into a position directly over the detector. The MCA reads information identifying the sample and its irradiation and counting parameters from a hard disk. This information is typically entered on the MCA keyboard prior to the chemical processing of the swabs (Figure 3). The MCA then acquires a spectrum and signals the sample changer to remove the current sample and bring in the next. The spectrum and its associated parameters are written out to the disk. This cycle is repeated until all samples have been counted.

When the gamma counting phase of the barium analysis is completed, the three standard spectra are recalled from the disk

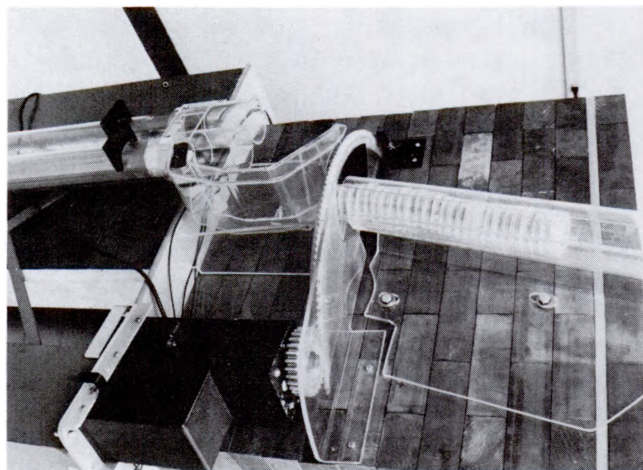


FIGURE 2. Sample changer loaded with petri dishes.

and the average activity at reactor discharge for 1 ug of barium is determined. This average activity is used to determine the micrograms of barium in each sample using the comparator method (Kruger 1971). The activity at reactor discharge is automatically calculated in each sample and compared to the average activity in the standards to determine the amount of barium in the sample. A computer printout, which contains sufficient information to manually determine this automatically calculated barium quantity, is produced for each sample (Figure 4). This "self-contained" printout is useful for inspection and review by the analyst prior to interpreting the meaning of the results and also serves as a permanent record of the analysis. As pointed out, the antimony results are obtained after reloading the sample changer and initiating the antimony counting and calculating routine (Figure 5).

An important requirement for the computer program used in the GSR analysis is that it make available the data points and the irradiation and counting parameters used to determine the antimony and barium concentrations. This is particularly evident

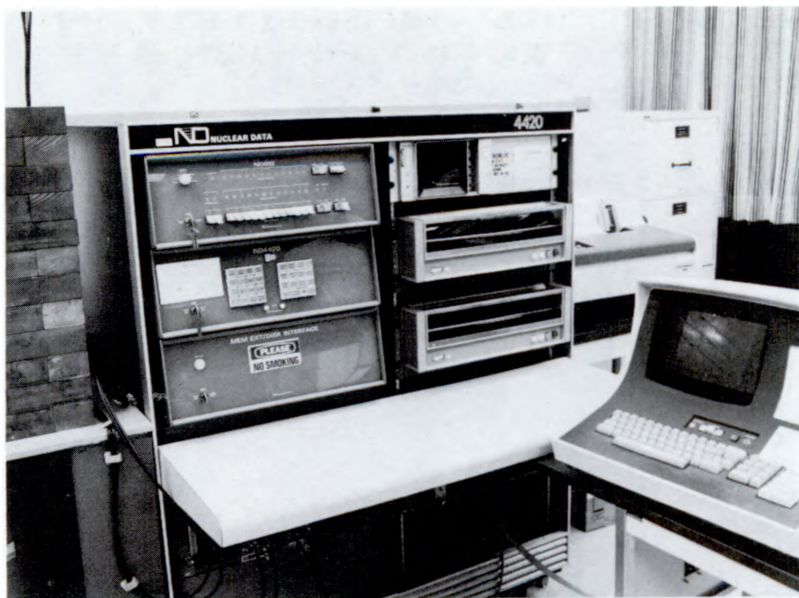


FIGURE 3. Multichannel analyzer and terminal.

in the peak integration routine. This routine does not determine if an actual photopeak is present. Instead, a region of interest is defined by calibrating the MCA carefully so that the photopeak of interest is present in its appropriate region. If there is no photopeak in a specific sample (this often happens for antimony, especially in the materials blank sample), the MCA will still calculate a value. Based on an inspection of the region on the printout and our experience as to the limit of detection for antimony and barium under the conditions of the analysis (approximately 20 ng for barium and 5 ng for antimony depending on presence or absence of contaminants and counting conditions), it is readily apparent when this automatically determined value is not meaningful. Some of the peak integration terms which appear in Figures 4 and 5 are defined in Figure 6.

INTERPRETATION OF DATA AND DISCUSSION

The FBI Laboratory examines GSR samples from the hands of approximately 2000 persons each year. This requires the application of batch processing methods, automatic gamma counting and automatic

data reduction. However, accurate quantitative data are only one factor, albeit an important one, considered by the analyst when making a judgment about the significance of the quantities of antimony and barium detected in an actual case. Other factors include the analyst's training and experience in GSR analysis and verifiable case information.

The FBI analyst has personally participated in activities which generate data from test firings and handblanks. Over 400 firearms have been test fired in the FBI Laboratory. Many of these firearms have been fired several times to determine the effect of factors such as environment, brand of ammunition, and intervening objects have on the deposition of GSR on the hands. In a majority of these test firings the range for barium is from 0.4 ug to 5.0 ug and for antimony 0.04 ug to 2.5 ug. Hundreds of handblanks have been collected and analyzed to determine amounts of antimony and barium which can be found on the hands of nonshooters.

The analyst should have information concerning the shooting and the swabbing of the suspected gunman; for instance, the type and caliber of firearm involved, the brand of ammunition, the number of shots, and the elapsed time between the shooting and the swabbing. Often, test firing the actual

**FBI LABORATORY WASHINGTON, D. C.
GUNSHOT RESIDUE ANALYSIS**

3/29/85 946:32

Q1 50305065 LK

ACTIVATION END: 3/27/85 10 3:00 RT-3 3 MIN.
ACQUISITION START: 3/27/85 1222:25 GELI 15A
PRESET TIME: 96 SEC
ELAPSED REAL TIME: 97 SEC
ELAPSED LIVE TIME: 96 SEC
CALIBRATION DATE: 3/27/85
KEY/CHNL: .500
OFFSET: 0.000

297	31	30	30	28	27	21	23
304	30	32	33	22	25	30	38
311	27	34	25	27	23	31	20
318	25	26	36	19	33	32	36
325	31	38	27	33	41	40	132
332	385	331	109	34	28	18	20
339	20	18	29	33	23	24	18
346	20	28	16	25	32	30	26
353	20	25	22	29	37	28	26
360	26	22	19	34	21	30	25

MICROGRAMS BARIUM 896

% RSD COUNTING 3.9

CNTS/UG BARIUM/MIN. 1916
CENTROID CHANNEL 332
HALF-LIFE BA-139 83 MIN.
BACKGROUND SPACING 20
BACKGROUND CHANNELS 10
PEAK CHANNELS 5

FIGURE 4. Computer printout of barium analysis.

**FBI LABORATORY WASHINGTON, D. C.
GUNSHOT RESIDUE ANALYSIS**

3/29/85 10 0:26

1 50305065 LK

TIVATION END: 3/27/85 10 3:00 RT-3 3 MIN.
 QUISITION START: 3/27/85 1516:21 GELI 15A
 ESET TIME: 900 SEC
 APSED REAL TIME: 911 SEC
 APSED LIVE TIME: 900 SEC
 LIBRATION DATE: 3/27/85
 KEY/CHNL: .500
 OFFSET: 0.000

1093	24	37	37	42	35	33	36
1100	31	29	35	35	33	39	49
1107	106	196	206	145	60	41	31
1114	32	46	44	52	51	40	37
1121	35	29	43	40	57	146	433
1128	678	516	194	74	34	21	25
1135	35	41	35	28	25	52	40
1142	41	34	39	34	48	36	29
1149	49	25	36	30	36	30	41
1156	30	32	38	28	42	34	47

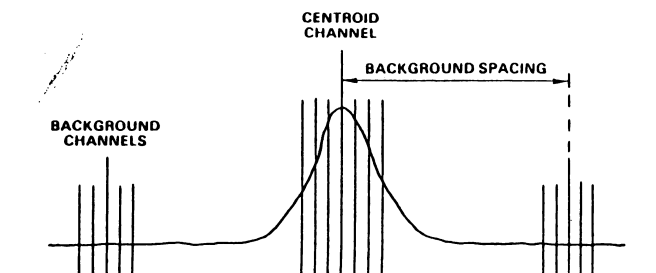
ICROGRAMS ANTIMONY 193

RSD COUNTING 2.5

TS/UG ANTIMONY/MIN. 677
 NTROID CHANNEL 1128
 LF-LIFE SB-122 4032 MIN.
 CKGROUND SPACING 30
 CKGROUND CHANNELS 10
 RK CHANNELS 7

FIGURE 5. Computer printout of antimony analysis.

PEAK INTEGRATION TERMS



CENTROID CHANNEL: Channel around which maximum peak area is determined. Channel entered or stated in program ± 2 channels

BACKGROUND CHANNELS: Sum of channels on both sides of peak

FIGURE 6. Definition of terms used in peak integration routine.

firearm involved in the case is very valuable.

The analyst must finally decide if the amounts of antimony and barium measured in the case under consideration can be associated with the discharge of a firearm or are such that no association can be made. If the analyst associates the antimony and barium with the discharge of a firearm, the opinion in most cases will be that the amounts of antimony and barium are typical of the amounts found on the hands of a person who has discharged a firearm or had his hands in close proximity to a discharging firearm. This latter clause is necessary, in our view, because it is possible to get GSR on the hands if one is near a discharging firearm held by another person. For instance, a struggle for a firearm or a shot at close range by an assailant can deposit GSR on the hands of the victim. In suicide investigations, the FBI Laboratory report generally points out that there is a possibility that the GSR could have been deposited if the victim was shot at close range by another person as well as through a self-inflicted situation.

If the amounts of antimony and barium cannot be associated with the discharge of a firearm, it is so reported; however, an explanation is added which advises the reader that this does not preclude the possibility that the person could have discharged a firearm and no GSR was deposited on the hand or that this residue, if deposited on the hand, was removed by

washing or wiping before the specimens were obtained.

The problem of the inconclusive examination is a difficult one. Among the possibilities for the absence of significant amounts of antimony or barium on the hands are that the person did not discharge a firearm, the firearm did not deposit GSR under the conditions of the shooting incident, the deposited GSR was removed by some activity prior to the swabbing, the swabbing was inadequate, or the chemical analysis was faulty (Kilty 1975). Of all these possibilities or combinations of them, only the last one is under the control of the analyst. It is not possible for the analyst to assign relative probabilities of occurrence for these factors; however, expert testimony is sometimes required so that the trier of fact will have available information concerning them.

The determination that there are GSR particles or chemical elements that are present in GSR on the hands does not prove that a person held and discharged a firearm. Nevertheless, it is still a valuable one. It at least places the person in an environment of GSR and can be a significant investigative aid in a shooting investigation.

ACKNOWLEDGMENT

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